

Estimating Fetal Gestational Age from Ultrasound Images using Deep Learning Techniques

N. Mohana Suganthi¹

Department of Computer Science and Engineering
Vel Tech Rangarajan Dr.Sagunthala
R&D Institute of Science and
Technology, Avadi, Chennai
mohanasuganthin@veltech.edu.in

Dr. R. Subhashni²

Department of Computer Applications
St.Peter's Institute of Higher
Education and Research, Avadi ,
Chennai -54
subhashniraj2018@gmail.com

A. Chitra³

Department of Computer Applications
St.Peter's Institute of Higher Education and
Research
prof.chitraa@gmail.com

S. Rajakumari⁴

Department of Computer Applications
St.Peter's Institute of Higher Education
and Research, Avadi ,
Chennai -54
grajiapk@gmail.com

Dr. T. Priya⁵

Department of Computer Application
EastPoint College of Higher
Education,
Jnana Prabha, East Point Campus,
Bengaluru, Karnataka 560049
suga.priya04@gmail.com

P. Caroline Anitha⁶

Department of Computer Applications
St.Peter's Institute of Higher Education and
Research
carolineanitha@gmail.com

Corresponding Author : mohanasuganthin@veltech.edu.in

Abstract—Monitoring fetal development and guaranteeing mother health depend on correct estimate of fetal gestational age. Conventional approaches of gestational age estimate mostly depend on manual estimations from ultrasonic scans, which may be labor-intensive and prone to human mistake. Variability in operators, picture quality, and fetal position might potentially affect current methods and cause possible errors and inconsistent outcomes. We offer the Ultrasound Images Using Deep Learning Techniques (UI-DLT) framework, a strong, automated method for fetal gestational age estimate, to handle these problems. UI-DLT analyzes ultrasonic pictures and enhances the accuracy and consistency of gestational age estimations by using deep learning—more especially, convolutional neural networks (CNNs). UI-DLT reduces human error and standards outcomes across several clinical environments by automating the analytic process. CNNs are used in the proposed system to categorize and examine ultrasonic pictures, therefore spotting important information on fetal development and growth. By extracting deep picture elements that match developmental phases, our method guarantees accurate and consistent gestational age estimations, thereby beyond the capacity of conventional manual techniques. Experimental data on ultrasonic image datasets shows significant increases in estimating accuracy of the UI-DLT framework. This approach not only improves the accuracy of gestational age prediction but also helps to progress clinical decision-making and mother-fetal health care.

Keywords—fetal gestational, Deep Learning, Ultrasound Images, monitoring, maternal health

I. INTRODUCTION

The correctness of the fetal gestational age (GA) computation provides both appropriate development of the baby and non compromise of the mother's health[1]. Moreover, this calculation is taken into account when deciding on healthcare choices that may include everything from prenatal tests to medications [2]. Usually, Georgian estimate is based on hand-made measurements employing ultrasonic images spanning

the biparietal diameter and crown-rump length [3]. The process revolves mostly on these measures[4]. These techniques might be time-consuming and prone to errors resulting from human contact as they depend on the knowledge and experience of the operator [5]. Apart from differences in hand measurements, fetal position, and imaging quality, additional factors might cause results that contradict one other and be erroneous [6].

Deep learning offers several interesting opportunities to be used to raise GA estimates [7]. These decisions have evolved because of presently available advancements[8]. A kind of deep learning model, convolutional neural networks (CNNs) have shown considerable potential in the field of medical imaging as they can learn intricate characteristics from vast datasets [9]. One kind of neural network are CNNs[10]. By lowering the quantity of mistakes that occur, CNNs for GA estimations enable standardizing measurements and automation, hence producing uniformity in a variety of healthcare environments [11].

As a way to estimate GA from ultrasonic images [12], we provide an automated CNN-based method based on the Deep Learning Techniques (UI-DLT) framework[13]. CNN forms the foundation of this technique[14]. This structure seeks to extract deep picture features compatible with the phases of fetal development [15]. This is carried out with the aim of surpassing present methods in terms of precision and dependability, hence improving accuracy[16]. Apart from more precise GA estimations, more consistent mother-fetal health care enhances clinical decision-making using the UI-DLT system, as the results of the studies [17] clearly reveal.

The goal of this work aims to minimize human error and variability in the process of determining the gestational age of the baby by employing an automated and standardized approach that generates advances in accuracy and consistency

across clinical environments. Reducing the degree of human error will help to achieve this.

Often used to characterize the two methods used in the process of estimating the gestational age based on ultrasonic imaging are operator-dependent variability and consistency. It addresses these problems by implying a deep learning-based method to provide accurate and consistent GA predictions, hence reducing human error.

Contribution of this paper

- Proposed an automated deep learning framework, UI-DLT, for precise fetal gestational age estimation using ultrasound images, reducing human error.
- Developed a CNN-based image classification approach that extracts key fetal features, improving accuracy over traditional manual measurement methods.
- Demonstrated significant accuracy improvements in gestational age prediction, enhancing clinical reliability and supporting informed maternal-fetal health management.

The remaining of this paper is structured as follows: In section 2, the related work of estimating fetal gestational is studied. In section 3, the proposed methodology of UI-DLT is explained. In section 4, the efficiency of UI-DLT is discussed and analyzed. Finally, in section 5 the paper is concluded with the future work.

II. LITERATURE SURVEY

Several artificial intelligence-based approaches for estimating fetal gestational age (GA) from ultrasonic images are investigated in this set of work. DeepGA, AGAEP, AI-BM-GAE, and POCUS-based AI GA Estimation Framework are among the methods meant to increase GA estimate accuracy in clinical settings particularly in low-resource environments where access to trained sonographers and expensive equipment is limited.

DeepGA

It build an automatic DeepGA model to attain entirely autonomous GA prediction in an end-to-end approach. Using a deep segmentation model (DeepSeg), this algorithm precisely detects and segment three vital tissues including the cranium, abdomen, and femur in which the shape is automatically extracted. It then can directly estimate the GA with a deep regression model (DeepReg). It assess DeepGA on a sizable dataset comprising 10,424 ultrasonic images from 7113 subjects. With a mean absolute estimate error (MAE) of 5 days, it shows better performance than the conventional measuring technique[18]. The DeepGA model is a new automatic solution based on artificial intelligence learning that can enable radiologists to improve the performance of GA estimation in several clinical settings, consequently increasing the efficiency of prenatal investigations.

Automatic Gestational Age Estimation Pipeline (AGAEP)

Clinical management of pregnancy depends critically on accurate evaluation of fetal Gestational Age (GA). Obstetric ultrasonic (US) technology is used in industrialized nations to estimate this. Using a low-cost obstetric US, automatic measurement of fetal structures in low- and middle-income

countries may help to establish GA without the need for experienced sonographers[19]. It describe a fully automated pipeline for image classification, segmentation, and structure measurement to estimate the GA and show a method for creating raw images suitable for model training by removing caliper and text annotation.

AI Brain Morphology GA Estimator(AI-BM-GAE)

Specifically in the transthalamic axial plane, the proposed method estimates gestational age (GA) by means of an artificial intelligence (AI) approach based on analysis of fetal brain morphology on conventional cranial ultrasound images[20]. Aiming to improve accuracy in the second and third trimesters when conventional approaches are less reliable, this automated AI model estimates GA by spotting patterns within fetal brain structure. Promising results were obtained from the model tested both alone as a tool and in conjunction with conventional fetal biometric criteria. In the third trimester, combining the AI model with biometric data enhanced GA estimation accuracy, reducing error rates compared to conventional biometrics alone.

POCUS-based AI GA Estimation Framework

Designed for point-of-care use in Low- and Middle-Income Countries (LMICs), this work presents a framework for using machine learning to estimate fetal gestational age (GA) from short fetal ultrasonic scans[21]. Two main obstacles are covered by the framework: the lack of reasonably priced, portable devices and the steep learning curve for ultrasonic mastery. It detects the head transcerebellar plane, gauges the transcerebellar diameter, and estimates GA using an inexpensive wireless ultrasonic probe, a smartphone or tablet, and automated image processing algorithms. Using conventional ultrasonic data, proof-of-concept analyses show the accuracy of these algorithms for clinical uses.

From ultrasonic images, automated gestational age estimates are provided by the proposed artificial intelligence-based systems DeepGA, AGAEP, AI-BM-GAE, and POCUS-based models. These models especially in the second and third trimesters improve accessibility and accuracy, so optimizing GA estimate in both low-income and developed environments.

Automatic early diagnosis of lung cancer

An automated system designed to assist in the early diagnosis of lung cancer by calculating the probability of malignancy and segmenting lung CT scan images is presented in [22]. The proposed approach enhances CT images using an enhancement filter and applies neural networks to reduce false positives in the identification of nodules. The system also calculates the likelihood of malignancy based on clinical and radiographic parameters, using Bayesian methods. Results show that the system can effectively identify pulmonary nodules and predict malignancy probability, which helps reduce the diagnostic burden on radiologists.

Machine learning based prediction

A machine learning-based method for improving the prediction of cardiovascular disease (CVD) is presented using an optimization algorithm called Gradient Descent Optimization (GDO) [23]. The optimization method improves the precision, sensitivity, and specificity of the CVD

prediction, making it a promising tool for timely diagnosis. The proposed model also outperforms existing techniques in terms of classification performance, indicating its potential for better healthcare applications in cardiovascular disease diagnosis.

III. PROPOSED METHOD

This set of figures demonstrates several ways for approximating fetal gestational age using ultrasonic images. It addresses deep learning models using CNNs and residual blocks as well as deep learning models including Elastic Net and Support Vector Regression, thus offering precise predictions for improved prenatal care.

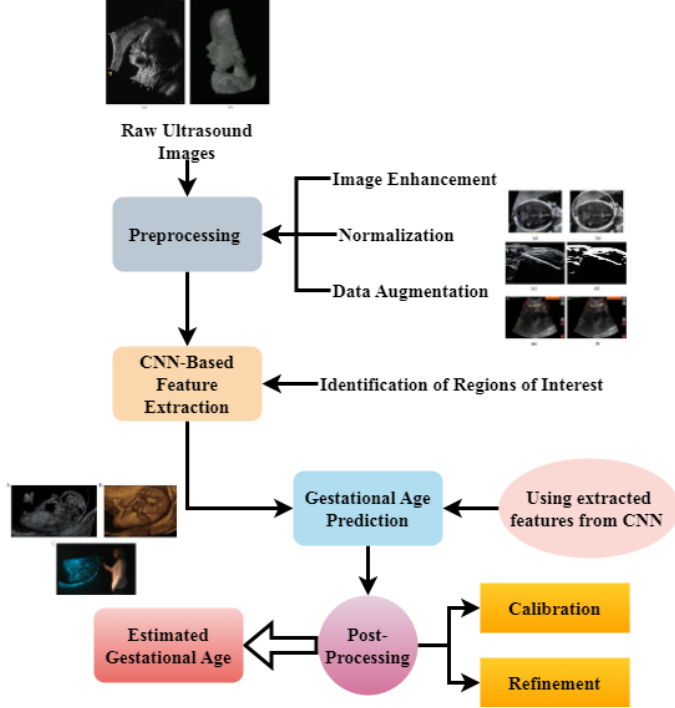


Fig. 1. Estimating Fetal Gestational Age Using Deep Learning from Ultrasound Images

Figure 1 shows how deep learning approaches might estimate fetal gestational age from ultrasonic images. Raw ultrasonic images start the process; preprocessing images enhances, normalizes, and generates data augmentation[24]. Key elements from a CNN-based model are extracted and then areas of interest including the fetal head and femur are identified. Following that, the model uses calibration and post-processing methods to polish the gestational age estimate. The expected gestational age is the end result, and it gives doctors important knowledge for suitable prenatal treatment.

$$\frac{e}{rt} * Q(l - uy'') = B(u) * M(l - pq'') + z(i - \partial v(l - wq''))(1)$$

The link $\frac{e}{rt}$ between the complex image features $Q(l - uy'')$ that are retrieved from the convolutional layers ($B(u)$) and their transformation z via feature mapping ($M(l - pq'')$) and adjustment parameters ($i - \partial v(l - wq'')$) is represented by Equation (1). This equation summarizes how different picture

modifications lead to consistent gestational age predictions. This is accomplished by modeling developmental markers.

$$\frac{f}{v(l - wq'')} : \rightarrow N(L - wqt'') + V(L - awt'') \quad (2)$$

A mapping function $\frac{f}{v(l - wq'')}$ is represented by Equation (2).

This function is responsible for guiding the translation of image features into outputs $N(L - wqt'')$ and $V(L - awt'')$,

which are indicative of predictive corrections and feature normalization in the CNN. It is important to note that the equation highlights the power of the approach to standardize predictions by applying adaptive modifications.

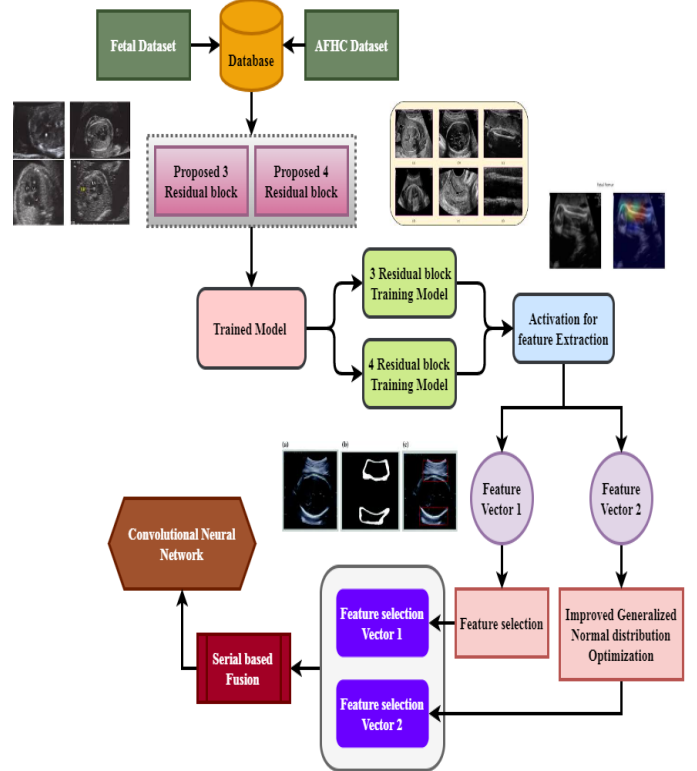


Figure 2 presents a deep learning-based approach for fetal gestational age estimate using ultrasonic images. First it gets datasets and runs two suggested residual block models—3-block and 4-block across these. Bayesian optimization helps one to fine-tune the hyperparameters. Improved optimisation methods enable the extraction and polish of features[25]. Two feature vectors are generated and merged serially for highest accuracy. After that, a neural network handles the fused features to project the gestational age. This framework mixes advanced image processing and optimization to generate consistent forecasts.

$$B(P - av') \equiv N(U - ab''), C(u - \partial t) \rightarrow D(u' - pa) \quad (3)$$

A relation that indicates the equivalence between feature changes in the convolutional network $N(U - ab'')$ and normalized output transformations are expressed by Equation (3). On the expression $B(P - av')$ reveals that progressive mapping of feature weights is necessary for refined

predictions with the help of this relation. This ensures that the estimation of the fetal gestational age is accurate by responding to the variations in the images.

$$2uv' > \partial n_2 - \{Qnj - IPM\} + J(U_2 A) \quad (4)$$

Equation (4) is associated with the UI-DLT framework $J(U_2 A)$, which involves the use of deep learning to handle

intricate picture feature interactions $2uv'$ and thresholds

$$\partial n_2 - \{Qnj - IPM\}, \text{ hence improving model predictions to}$$

improve accuracy. The objective is to make certain that the feature analysis of the framework exceeds the criteria for noise and variability.

$$4(LV' - qam'(y * dz)), Y_t(e * pki' + qa) \quad (5)$$

As a result of the correlation between Equation (5) and how deep learning models adapt $y * dz$ and scale retrieved characteristics $4LV' - qam'$, the accuracy Y_t of predictions of fetal gestational age is improved $pki' + qa$. The objective of this equation is to illustrate the framework's capacity to provide specific and trustworthy outcomes.

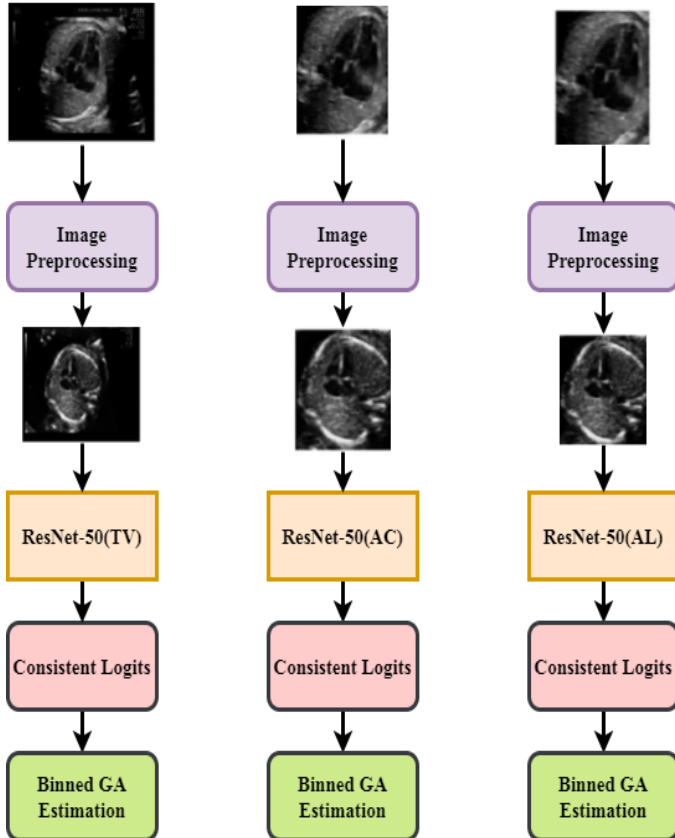


Fig. 3. Flow of Image preprocessing

Figure 3 uses ultrasonic images to demonstrate a deep learning technique for fetal gestational age estimation. Three parallel pipelines handling different fetal biometric views Here involved are transverse view (TV), abdominal circumference (AC), and femur length (FL). Each image is preprocessed to increase quality before being fed into various ResNet-50 models catered for every view[28]. The models produce consistent logits that are then used for binned gestational age estimate, so extracting key features. This framework combines several fetal measurements to provide accurate and consistent age predictions, so supporting prenatal care and assessment.

$$lv''(n - ap'') \rightarrow e_v Lp(qwj - xz'') + laq'' \quad (6)$$

Equation (6) provides a visual representation of the transformation of features by means of weighted adjustments $(lv''(n - ap''))$ and parameter updates $(e_v Lp)$. This is then

followed by the computation of refined outputs laq'' via

further processing $(qwj - xz'')$. The objective of the

equation is to capture the continual optimization of feature extraction and output prediction.

$$H(Sp - avn'') \rightarrow \alpha_2 Ab(S(\forall \equiv rq'')) \quad (7)$$

The transformation that is indicated by Equation (7) involves the feature set $S(\forall \equiv rq'')$ being subjected to adjustments $(H(Sp - avn''))$ and being mapped to a refined output (Ab) depending on a particular condition $(\alpha_2 Ab)$. This is done to ensure that the model is capable of processing different inputs in an adaptable manner and producing accurate and standardized predictions on the analysis of human error.

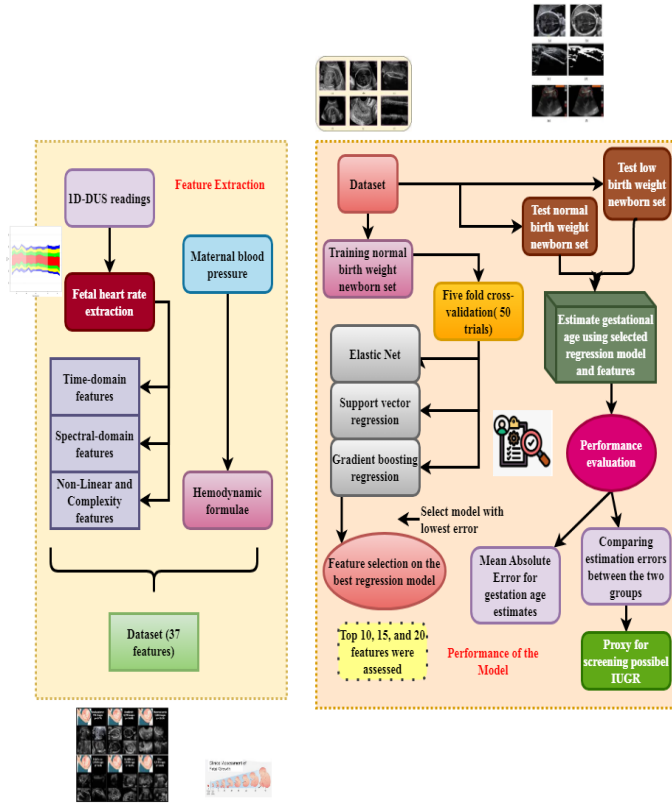


Fig. 4. Predicting Fetal Gestational Age

Designed to ascertain the gestational age of the baby, Figure 4 shows a standard deep learning pipeline built. Ultrasounds scans allow one to gather important data such the mother's blood pressure, the heart rate of the fetus, and the features of blood flow. Feeding these features into deep learning models such as Elastic Net, Support Vector Regression (SVR), and Gradient Boosting Regression (GBR) helps the models to provide forecasts about the gestational age of probable babies. The dataset is split five times using the five-fold cross-validation method for training and testing. This ensures that the assessment of the model is quite exact[27]. Often referred to as mean absolute error, or MAE, this statistic evaluates the average deviation between the expected and actual gestational ages. This provides a reasonable gauge of the accuracy of the prognosis. The figures show advanced methods for estimating fetal gestational age including deep learning models using image processing, feature extraction, and optimization techniques. These techniques increase the accuracy of gestational age prediction, thus guiding doctors in their prenatal therapy and assessment decisions.

$$E - vt(L - qat'') \rightarrow Xz(L - paw'') \quad (8)$$

Equation (8), in which the input features $E - vt$ are subjected to transformations $(L - qat'')$ and are transferred to refined

outputs (Xz) , thus capturing important characteristics

$L - paw''$ of the data. The objective of this equation is to

describe the process by which the system improves picture feature extraction and prediction mapping for the analysis of clinical reliability.

$$Up(l - aq') \rightarrow Naq(L - amt'') + Cqa'' \quad (9)$$

Equation (9), which serves as a representation $L - amt''$ of

the transformation of features (Up) via adjustments and

mappings Cqa'' , results in improved outputs $(l - aq')$ and

further modifications (Naq) . The objective of this equation is

to describe the iterative refinement of feature extraction and output prediction on the analysis of estimation accuracy.

$$H(Lak - zb:nl') \rightarrow \partial \propto [Pw - 2vf''] \quad (10)$$

Equation (10), which has been converted H and altered $Pw - 2vf''$ to give refined outputs $Lak - zb:nl'$, which include certain weightings $\partial \propto$ and scaling factors. This equation's objective is to offer a description of the final modifications that have been made to the image characteristics for the analysis of gestational age prediction.

IV. RESULTS AND DISCUSSION

While precise assessment of fetal gestational age determines prenatal therapy, human error and operator variability abound in current manual approaches. The UI-DLT system uses deep learning more particularly, CNNs to automatically analyze ultrasonic images, therefore generating consistent, accurate gestational age estimates in clinical environments.

This dataset presents a Computer-Aided Detection (CAD) approach for automated fetal Head Circumference (HC) measurement in 2D ultrasound images for all trimesters. HCs can measure gestational age and fetal development. Developing countries with sonographer shortages may benefit from automated HC assessment. The CAD system comprises two steps: The fetus skull was found utilizing Haar-like features in a random forest classifier trained on ultrasound images. Second, Hough transform, dynamic programming, and ellipse fit found HC. After training on 999 photographs, 335 shots from all trimesters independently tested the CAD system. A skilled sonographer and medical researcher marked the test set. Reference gestational age was calculated using CRL. Findings show CAD acts like a skilled sonographer. System results are equivalent or better than literature-

published systems. The first automated HC assessment system with complete trimester data was evaluated on a large test group.

A. Analysis of human error (low)



Fig. 5. Graphical Representation of Human Error Analysis

Employing which operator expertise is greatly minimized, the proposed UI-DLT design reduces reliance on manual measurements which are prone to operator skill and experience variances (figure 5). With human error at a low level of 29.41%, the automated method regularly evaluates ultrasonic pictures across different clinical contexts to standardize gestational age estimates. Reducing human error increases the accuracy of measurements and helps minimize variations resulting from various points of view, strengthening the estimate of fetal age.

B. Analysis of clinical reliability

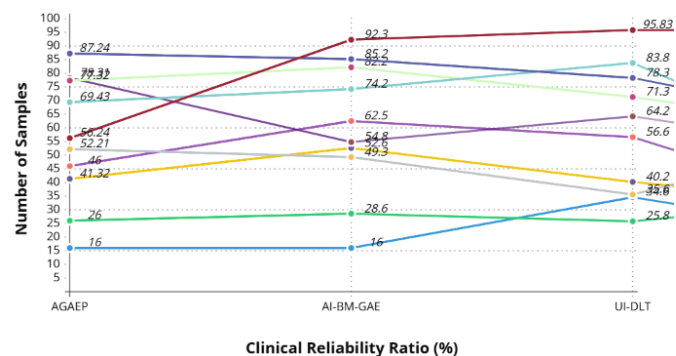


Fig. 6. Graphical Representation of clinical reliability analysis

With strong clinical dependability that is, persistence in producing consistent gestational age estimates throughout numerous patient events and imaging environments UI-DLT displays at 95.83% (Figure 6). Automating feature extraction and analysis enables the system to lower generally occurring variability related to different ultrasonic operators, image quality, and fetal placement. This consistency helps to build confidence in the accuracy of clinical judgments and assures consistent prenatal treatment throughout several healthcare settings.

C. Analysis of estimation accuracy

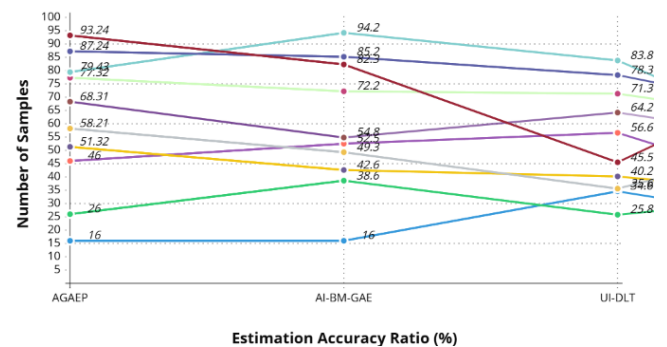


Fig. 7. Graphical Representation of estimation accuracy analysis
With 93.81% estimate accuracy, UI-DLT is quite accurate. One may get this degree of precision by use of convolutional neural networks identifying and analyzing complex visual features connected with phases of fetal development (Figure 7.). The deep learning model not just learns tiny traits that traditional methods overlook but also handles huge datasets, enhancing prediction accuracy. Improved patient care results from reduced probable misclassifications of gestational age brought about by increasing accuracy.

D. Analysis of gestational age prediction

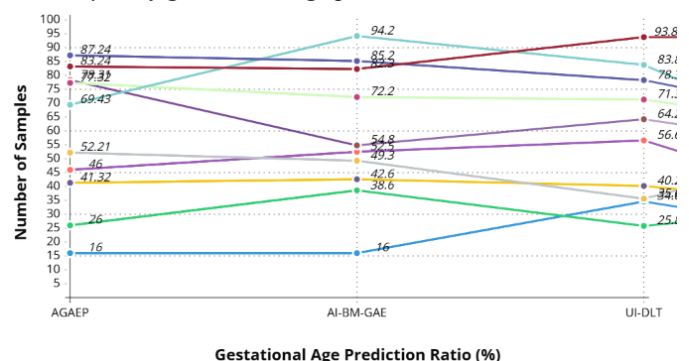


Fig. 8. Graphical Representation of gestational age prediction analysis
UI-DLT has a 98.22% success rate and a very accurate gestational age prediction. By employing CNNs to analyze significant fetal features from ultrasonic images, the method provides correct developmental stage predictions, therefore enabling quick therapeutic interventions when required (figure 8). This prediction accuracy allows clinicians to make informed decisions for prenatal care and reduces the risks associated with inaccurate gestational age estimation, thereby managing mother-fetal health.
UI-DLT reduces human error and increases clinical dependability by determining gestational age with remarkable accuracy (98.22%), by automating image processing over CNNs. Excellent performance across many patient cases enables the framework to provide consistent, reliable prenatal assessments, directing clinicians to make correct, informed decisions for mother-fetal health care.

V. CONCLUSION

Automating the analysis of ultrasonic images using the Ultrasound Images Using Deep Learning Techniques (UI-

DLT) framework greatly improves the estimate of fetal gestational age. UI-DLT improves accuracy and consistency in gestational age prediction using convolutional neural networks (CNNs), thus overcoming sometimes influenced by operator variability, image quality, and fetal position conventional manual techniques. With a 98.22% success rate in gestational age prediction, a 93.81% estimate accuracy, and a clear drop in human error to 29.41%, experimental data show that UI-DLT achieves great accuracy. Moreover, the framework shows great clinical dependability at 95.83%, proving its resilience in many clinical settings and over trimesters. Standardizing outcomes across many healthcare facilities guarantees consistent results that, employing consistency, can support clinical decision-making and mother-fetal health management. UI-DLT reduces reliance on human knowledge and manual measurement, offering a consistent, easily available tool for fetal monitoring, especially in areas with limited access to certified sonographers. Future work on UI-DLT could concentrate on improving its generalizability and robustness by including more varied ultrasonic datasets over several populations and clinical settings. Moreover, improving the system to automatically measure other fetal biometrics, such head circumference and femur length, helps support thorough fetal growth monitoring. Creating a lightweight model for cellphones or portable ultrasonic devices might help make the technology more widely available in places with limited resources. Including explainability elements into the framework will help doctors better grasp the model's choices, hence boosting trust in AI-driven gestational age predictions in clinical environments.

REFERENCES

1. Lee, L. H., Bradburn, E., Craik, R., Yaqub, M., Norris, S. A., Ismail, L. C., ... & Papageorgiou, A. T. (2023). Machine learning for accurate estimation of fetal gestational age based on ultrasound images. *NPJ digital medicine*, 6(1), 36.
2. Lee, C., Willis, A., Chen, C., Sieniek, M., Watters, A., Stetson, B., ... & Gomes, R. G. (2023). Development of a machine learning model for sonographic assessment of gestational age. *JAMA network open*, 6(1), e2248685-e2248685.
3. Fung, R., Villar, J., Dashti, A., Ismail, L. C., Staines-Urias, E., Ohuma, E. O., ... & Miller, R. (2020). Achieving accurate estimates of fetal gestational age and personalised predictions of fetal growth based on data from an international prospective cohort study: a population-based machine learning study. *The Lancet Digital Health*, 2(7), e368-e375.
4. Kojita, Y., Matsuo, H., Kanda, T., Nishio, M., Sofue, K., Nogami, M., ... & Murakami, T. (2021). Deep learning model for predicting gestational age after the first trimester using fetal MRI. *European Radiology*, 31, 3775-3782.
5. Ananth, C. V., & Brandt, J. S. (2020). Fetal growth and gestational age prediction by machine learning. *The Lancet Digital Health*, 2(7), e336-e337.
6. Fiorentino, M. C., Villani, F. P., Di Cosmo, M., Frontoni, E., & Moccia, S. (2023). A review on deep-learning algorithms for fetal ultrasound-image analysis. *Medical image analysis*, 83, 102629.
7. Ramirez Zegarra, R., & Ghi, T. (2023). Use of artificial intelligence and deep learning in fetal ultrasound imaging. *Ultrasound in Obstetrics & Gynecology*, 62(2), 185-194.
8. Shen, L., Zheng, J., Lee, E. H., Shpanskaya, K., McKenna, E. S., Atluri, M. G., ... & Yeom, K. W. (2022). Attention-guided deep learning for gestational age prediction using fetal brain MRI. *Scientific reports*, 12(1), 1408.
9. Mishra, D., Chandramohan, V., Sharma, N., Gosain, M., Wadhwa, N., Natchu, U. C. M., ... & Bhatnagar, S. (2024). Development and external validation of an ultrasound image-based deep learning model to estimate gestational age in the second and third trimesters of pregnancy using data from Garbh-Ini cohort: a prospective cohort study in North Indian population. *medRxiv*, 2024-05.
10. Cengiz, S., & Yaqub, M. (2021, September). Automatic fetal gestational age estimation from first trimester scans. In *International Workshop on Advances in Simplifying Medical Ultrasound* (pp. 220-227). Cham: Springer International Publishing.
11. Lu, Y., Fu, X., Chen, F., & Wong, K. K. (2020). Prediction of fetal weight at varying gestational age in the absence of ultrasound examination using ensemble learning. *Artificial intelligence in medicine*, 102, 101748.
12. Tran, C. B. N., Nedelec, P., Weiss, D. A., Rudie, J. D., Kini, L., Sugrue, L. P., ... & Rauschecker, A. M. (2023). Development of Gestational Age-Based Fetal Brain and Intracranial Volume Reference Norms Using Deep Learning. *American Journal of Neuroradiology*, 44(1), 82-90.
13. Lee, L. H., Bradburn, E., Papageorgiou, A. T., & Noble, J. A. (2020, October). Calibrated bayesian neural networks to estimate gestational age and its uncertainty on fetal brain ultrasound images. In *International Workshop on Advances in Simplifying Medical Ultrasound* (pp. 13-22). Cham: Springer International Publishing.
14. Sivari, E., Civelek, Z., & Sahin, S. (2024). Determination and classification of fetal sex on ultrasound images with deep learning. *Expert Systems with Applications*, 240, 122508.
15. Xia, T. H., Tan, M., Li, J. H., Wang, J. J., Wu, Q. Q., & Kong, D. X. (2021). Establish a normal fetal lung gestational age grading model and explore the potential value of deep learning algorithms in fetal lung maturity evaluation. *Chinese Medical Journal*, 134(15), 1828-1837.
16. Plotka, S., Klasa, A., Lisowska, A., Seliga-Siwecka, J., Lipa, M., Trzciński, T., & Sitek, A. (2022). Deep learning fetal ultrasound video model match human observers in biometric measurements. *Physics in Medicine & Biology*, 67(4), 045013.
17. Pei, Y., E, L., Dai, C., Han, J., Wang, H., & Liang, H. (2023). Combining deep learning and intelligent biometry to extract ultrasound standard planes and assess early gestational weeks. *European Radiology*, 33(12), 9390-9400.
18. Dan, T., Chen, X., He, M., Guo, H., He, X., Chen, J., ... & Cai, H. (2023). DeepGA for automatically estimating fetal gestational age through ultrasound imaging. *Artificial Intelligence in Medicine*, 135, 102453.
19. Prieto, J. C., Shah, H., Rosenbaum, A. J., Jiang, X., Musonda, P., Price, J. T., ... & Stringer, J. S. (2021, February). An automated framework for image classification and segmentation of fetal ultrasound images for gestational age estimation. In *Medical Imaging 2021: Image Processing* (Vol. 11596, pp. 453-462). SPIE.
20. Burgos-Artizzu, X. P., Coronado-Gutiérrez, D., Valenzuela-Alcaraz, B., Vellvé, K., Eixarch, E., Crispi, F., ... & Gratacos, E. (2021). Analysis of maturation features in fetal brain ultrasound via artificial intelligence for the estimation of gestational age. *American Journal of Obstetrics & Gynecology MFM*, 3(6), 100462.
21. Maraci, M. A., Yaqub, M., Craik, R., Beriwal, S., Self, A., Von Dadelszen, P., ... & Noble, J. A. (2020). Toward point-of-care ultrasound estimation of fetal gestational age from the trans-cerebellar diameter using CNN-based ultrasound image analysis. *Journal of Medical Imaging*, 7(1), 014501-014501.

22. Manoharan, S., 2020. Early diagnosis of lung cancer with probability of malignancy calculation and automatic segmentation of lung CT scan images. *Journal of Innovative Image Processing (JIIP)*, 2(04), pp.175-186.
23. Jebakumar Immanuel D; Sathesh Abraham Leo E; Saranya S; Nithiya C; Arunkumar K; Pradeep D. "An Intelligent Heart Disease Prediction by Machine Learning Using Optimization Algorithm", *Journal of Information Technology Management*, 16, 1, 2024, 167-181. doi: 10.22059/jitm.2024.96381
24. Torres, H. R., Morais, P., Oliveira, B., Birdir, C., Rüdiger, M., Fonseca, J. C., & Vilaça, J. L. (2022). A review of image processing methods for fetal head and brain analysis in ultrasound images. *Computer methods and programs in biomedicine*, 215, 106629.
25. Wang, Q., Liu, D., & Liu, G. (2022). [Retracted] Value of Ultrasonic Image Features in Diagnosis of Perinatal Outcomes of Severe Preeclampsia on account of Deep Learning Algorithm. *Computational and Mathematical Methods in Medicine*, 2022(1), 4010339.
26. Wyburd, M. K., Hesse, L. S., Aliasi, M., Jenkinson, M., Papageorgiou, A. T., Haak, M. C., & Namburete, A. I. (2021). Assessment of regional cortical development through fissure based gestational age estimation in 3D fetal ultrasound. In *Uncertainty for Safe Utilization of Machine Learning in Medical Imaging, and Perinatal Imaging, Placental and Preterm Image Analysis: 3rd International Workshop, UNSURE 2021, and 6th International Workshop, PIPPI 2021, Held in Conjunction with MICCAI 2021, Strasbourg, France, October 1, 2021, Proceedings 3* (pp. 242-252). Springer International Publishing.
27. Liu, L., Tang, D., Li, X., & Ouyang, Y. (2024). Automatic fetal ultrasound image segmentation of first trimester for measuring biometric parameters based on deep learning. *Multimedia Tools and Applications*, 83(9), 27283-27304.
28. <https://datasetsearch.research.google.com/search?src=0&query=Estimating%20Fetal%20Gestational%20Age%20from%20Ultrasound%20Images%20&docid=L2cvMTFwNG5fN2Jjcg%3D%3D>